

Macguire McDuff

+1 (304)-494-5448 | macguiremcduff@gmail.com | macguirevan.github.io | github.com/macguirevan

EDUCATION

University of Florida

August 2021 - May 2025

Bachelor of Science in Computer Science

Gainesville, FL

- GPA: 3.4; Dean's List (multiple semesters) and SEC Athletic Honor Roll (all semesters)
- Key Classes: Operations Systems, Enterprise Software Engineering, Machine Learning, and Database Systems

NCAA Swimmer

- 5 Time NCAA Champion, 12 Time NCAA All-American, and Current NCAA Record Holder

PROFESSIONAL EXPERIENCE

Associate Software Developer

June 2025 – Present

iPipeline

Fort Lauderdale, Florida

- Lead the development of an AWS Bedrock token vending CLI tool using Python, AWS Lambda, and Terraform, enabling 12-hour access to Claude AI for developers, with integrated cost management controls through AWS Cost Explorer.
- Developed and deployed Amazon Q business instances that seamlessly integrated with company support notebooks and Salesforce, improving information retrieval processes and significantly enhancing support and sales team productivity.
- Designed and implemented AWS CloudWatch alarms to monitor and govern internal usage of Claude AI code across the company

Software Engineer Intern

May 2023 – July 2023

Advantus Corp

Jacksonville, FL

- Collaborated within a small, agile team of developers to optimize and maintain proprietary systems, utilizing PHP to deliver scalable solutions.
- Performed maintenance and enhancement tasks on Oracle database platforms.

PROJECTS

RuneScape Market Analyzer | *React, Python, Flask, SQLite*

[GitHub](#)

- Built key frontend components in React based on UI/UX specifications, including interactive pages for viewing item data and trade analysis.
- Integrated dynamic data visualizations using Chart.js and managed routing and date formatting with react-router-dom and date-fns.
- Performed full-system testing and debugging across the frontend and backend, ensuring stability and feature compliance with project requirements.

ML Student Performance Predictor | *Python, Pandas, NumPy, scikit-learn, TensorFlow*

[GitHub](#)

- Led a team of four to develop machine learning models for predicting student exam performance, utilizing Python-based tools like scikit-learn, TensorFlow, and pandas for data processing, modeling, and evaluation.
- Conducted regression and classification tasks to predict numerical scores and pass/fail outcomes, leveraging metrics such as MSE, R2, accuracy, and F1 score to assess model performance, achieving top results of 97.03% accuracy.
- Explored 20 key student performance factors (e.g., hours studied, attendance, parental involvement) and implemented data preprocessing techniques on 6,607 records to ensure robust insights for educational support.

OTHER EXPERIENCE

USA National Team Member

April 2024 – July 2024

United States Olympic & Paralympic Committee

Colorado Springs, CO

Lifeguard

Summers 2018/2019

Dupont Employees Recreation Club Incorporated

Parkersburg, WV

SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS

Frameworks & Tools: React, Git, Docker, Node.js, Angular, TensorFlow, scikit-learn, Linux, SQL

Cloud & DevOps: AWS (Lambda, S3, API Gateway, Bedrock, Cloudwatch), Terraform, Terragrunt